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Injuries in Water Sports: A Structured Narrative Review of Epidemiology, Injury Patterns and Prevention Strategies

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Abstract

Introduction. Water sports include a wide range of recreational and competitive activities, such as swimming, surfing, windsurfing, kitesurfing, wingfoiling and sailing. Although these sports share the aquatic environment, they differ substantially in biomechanics, equipment use, speed and environmental exposure. As a result, they are associated with diverse injury patterns, ranging from overuse conditions to acute traumatic injuries.

Aim of the study. The aim of this structured narrative review was to summarize the available literature on injuries occurring in water sports, with particular emphasis on epidemiology, anatomical injury patterns, mechanisms of injury and prevention strategies across swimming, surfing, windsurfing, kitesurfing, wingfoiling and sailing.

Methods. A structured narrative review was conducted using PubMed/MEDLINE as the primary database. Additional relevant articles were identified through manual screening of reference lists. Articles were included if they addressed sports-related injuries in non-motorized water sports and provided information on injury epidemiology, anatomical location, mechanisms, risk factors, treatment, return to sport or prevention. A total of 25 articles were included and synthesized descriptively.

Conclusions. Water sports should not be considered a uniform injury category, as each discipline has a distinct injury profile. Prevention strategies should be sport-specific and include load management, technical training, physical conditioning, environmental risk assessment, safe equipment use and early recognition of symptoms. Further studies using standardized injury definitions and prospective surveillance are needed, particularly for newer disciplines such as wingfoiling and foiling-related windsurfing classes.

Keywords: water sports, sports injuries, swimming injuries, surfing injuries, windsurfing, kitesurfing, wingfoiling, sailing, injury prevention, overuse injuries

Introduction

Water sports represent a diverse group of recreational and competitive activities that include swimming, surfing, windsurfing, kitesurfing, wingfoiling, foiling-related sailing disciplines and traditional sailing. Although these sports share the aquatic environment, they differ substantially in terms of biomechanics, equipment use, speed, environmental exposure and injury mechanisms. As a result, water sports are associated with a broad spectrum of injuries, ranging from chronic overuse conditions to acute traumatic events related to falls, collisions, waves, wind, equipment and contact with the water surface or seabed [1,2].

Swimming is distinct from many other water sports because it is primarily characterized by repetitive movement patterns and high training volume. Consequently, swimming-related injuries are frequently associated with overuse mechanisms, particularly affecting the shoulder, lumbar spine and upper extremity. Previous reviews and epidemiological studies have emphasized the importance of training load, stroke mechanics, mobility, strength deficits and prevention strategies in competitive swimmers [3–8]. In contrast, board-based and wind-dependent water sports, such as surfing, windsurfing, kitesurfing and wingfoiling, are more often associated with acute injuries caused by falls, jumps, loss of control, contact with the board, sail, kite or foil, and changing environmental conditions [9–21].

Surfing injuries commonly involve acute trauma, including lacerations, head and facial injuries, and injuries to the extremities, while overuse complaints may also occur due to

paddling and prolonged prone positioning [9–12]. Kitesurfing and windsurfing have specific risk profiles related to wind force, speed, jumps, landings and equipment-related mechanisms, with several studies reporting orthopaedic injuries involving the lower extremities, trunk and upper extremities [13–19]. More recent disciplines, including wingfoiling and iQFOiL, remain less extensively studied, but the available literature suggests that hydrofoil equipment, high-speed falls and technical demands may create distinct injury patterns that require further investigation [2,20,21].

Sailing also represents a heterogeneous category, ranging from recreational offshore sailing to Olympic and high-performance racing. Injuries in sailing may result from falls on deck, contact with ropes, booms and other equipment, high-speed maneuvers, repetitive loading and prolonged exposure to environmental conditions. Existing studies suggest that injury patterns differ according to the type of sailing, level of participation and environmental context [22–25].

Despite the growing body of literature on individual water sports, the available evidence remains heterogeneous, with differences in study design, injury definitions, populations, exposure measurement and reporting methods. This makes direct comparison between disciplines difficult. However, a structured synthesis of the available literature may help identify common injury patterns, sport-specific mechanisms and potential prevention strategies.

Materials and methods

This study was designed as a structured narrative review of the literature concerning injuries occurring during water sports. The review focused on non-motorized recreational and competitive water sports, including swimming, surfing, windsurfing, kitesurfing, kiteboarding, wingfoiling, foiling-related windsurfing disciplines, iQFOiL and sailing.

A literature search was performed primarily using the PubMed/MEDLINE database. The search was supplemented by manual screening of the reference lists of selected articles in order to identify additional relevant publications. The final search was conducted on 12 June 2026. The search strategy included combinations of sport-specific and injury-related keywords, including terms related to swimming, surfing, windsurfing, kitesurfing, wingfoiling, sailing, injury epidemiology, overuse injuries, acute injuries and prevention.

Articles were considered eligible if they focused on injuries associated with the selected water sports and provided clinically relevant information on at least one of the following topics: injury epidemiology, anatomical injury location, injury type, injury mechanism, risk factors,

treatment, return to sport or prevention strategies. Studies involving recreational, amateur, competitive and elite athletes were considered eligible. Original studies, epidemiological studies, prospective and retrospective observational studies, cross-sectional surveys, systematic reviews, narrative reviews and clinically relevant review articles were included.

Articles were excluded if they did not focus on sports-related injuries in water sports, if they concerned primarily motorized water sports, or if they focused exclusively on drowning, near-drowning, hypothermia, infections, dermatological conditions, marine animal injuries, scuba diving or diving-related barotrauma without relevant information on musculoskeletal or sports-related injuries. Case reports, letters to the editor, conference abstracts, editorials and non-peer-reviewed publications were excluded. Only articles published in English and available as abstracts or full texts were considered.

Based on these criteria, 25 articles were included in the final narrative synthesis. The included publications covered swimming, surfing, windsurfing, kitesurfing, wingfoiling, iQFOiL, sailing and broader aquatic or board-related orthopaedic injuries. Data were extracted descriptively from each article, including author, year of publication, sport discipline, study type, population, injury type, anatomical location, injury mechanism, risk factors and prevention-related findings.

Because of the heterogeneity of the included studies, including differences in study design, population characteristics, injury definitions and outcome reporting, no meta-analysis or statistical pooling was performed. The findings were synthesized narratively and organized according to sport discipline. The results were then compared across disciplines to identify common injury patterns, sport-specific mechanisms and potential prevention strategies.

Results

The included literature showed that injury patterns in water sports vary substantially depending on the type of activity, level of participation, equipment use and environmental exposure. Swimming was mainly associated with overuse injuries related to repetitive movement patterns and high training volume. In contrast, surfing, windsurfing, kitesurfing, wingfoiling and sailing were more frequently associated with acute traumatic injuries caused by falls, collisions, contact with equipment, wind, waves and variable environmental conditions. Across the analyzed sports, the most commonly discussed anatomical regions included the shoulder, spine, head and neck, trunk and lower extremities. The following sections summarize the main injury patterns reported for each water sport.

Swimming

Swimming differs from many other water sports because most injuries are related to repetitive loading rather than direct external trauma. Competitive swimmers are exposed to high training volumes and repeated stroke cycles, which place cumulative stress on the shoulder complex, spine and lower extremities. The shoulder is one of the most frequently discussed anatomical regions in swimming-related literature, reflecting the repetitive overhead nature of the sport and the high mechanical demand placed on the shoulder joint, scapular stabilizers and surrounding soft tissues [4,6–8].

Shoulder pain and upper extremity injuries in swimmers are generally considered multifactorial. Training volume, stroke mechanics, fatigue, muscle imbalance, reduced strength endurance, scapular control and inadequate recovery have all been discussed as potential contributors to symptoms [4,6–8]. A systematic review on shoulder pain and injury risk factors in competitive swimmers emphasized that evidence for many proposed risk factors remains limited or inconsistent, although workload-related factors and reduced posterior shoulder strength endurance may be clinically relevant [8]. This suggests that shoulder injury prevention should not rely on a single factor, but rather on a comprehensive approach addressing training load, technique, strength, mobility and recovery.

In addition to shoulder disorders, swimming-related injuries may involve the lumbar spine and lower extremities. Low back pain in competitive swimmers has been associated with swimming biomechanics, repeated lumbar extension and rotation, and stroke-specific movement patterns, particularly in athletes exposed to intensive training loads [3]. Lower limb complaints may occur in relation to kicking mechanics, breaststroke technique and repetitive loading during training [4,6]. Overall, the available literature indicates that swimming injuries are predominantly overuse-related and differ from the injury profile observed in board-based and wind-dependent water sports. Prevention strategies should therefore focus on load management, stroke technique, shoulder and trunk strengthening, mobility, adequate recovery and early response to pain before symptoms become persistent [3–8].

Surfing

Surfing is associated with both acute traumatic injuries and overuse complaints, but acute trauma is particularly prominent in the available literature. Unlike swimming, injury mechanisms in surfing frequently involve external forces, including wave impact, falls, collision with the surfboard, contact with fins, rocks, coral or sand, and interactions with other surfers [9–12]. Acute surfing injuries most often include lacerations, contusions, sprains,

fractures, dislocations and trauma to the head, face or neck [9–12]. These patterns are closely related to the specific exposure during paddling, take-off, wave riding and falling, when the surfer may collide with the board, fins, seabed, reef or other surfers [10–12]. Lower extremity injuries are also common and usually result from twisting mechanisms, direct impact or attempts to maintain balance during wave riding [10–12]. A systematic review of acute surfing injuries confirmed that skin injuries, board-related trauma, head, face and neck injuries, and lower limb involvement are among the recurring injury patterns reported in the literature [12]. Reviews have also indicated that being struck by one's own board or another surfer's board is among the most common mechanisms of injury in surfing [10,11].

Although acute injuries dominate, surfing may also be associated with overuse symptoms. Repetitive paddling, prolonged prone positioning, repeated lumbar extension and rotational demands may contribute to shoulder, cervical spine and lumbar spine complaints, particularly in surfers with high exposure or inadequate conditioning [10,11]. Therefore, prevention in surfing should address both traumatic and gradual-onset mechanisms. Suggested preventive strategies include education regarding wave conditions, crowding and environmental hazards, safe board handling, appropriate equipment, physical preparation, fatigue management and consideration of protective equipment in selected high-risk settings [10–12].

Windsurfing

Windsurfing combines board control, sail handling, balance, wind exposure and variable water conditions. As a result, its injury profile includes both acute traumatic injuries and overuse complaints. Acute injuries are commonly related to falls, loss of balance, collision with equipment, sudden changes in wind force and direct impact with the board, sail or water surface [13–15]. Injury mechanisms may vary according to skill level, weather conditions, equipment type and the maneuvers performed.

The literature on recreational windsurfing emphasizes that injury reporting is limited by inconsistent definitions and heterogeneous study designs, which makes comparison between studies difficult [13]. Reported injuries may involve the lower extremities, upper extremities, trunk, head and spine [13–15]. Falls during high-speed movement, failed maneuvers and equipment-related impact are important mechanisms of acute injury. At the same time, sustained stance, repetitive sail handling, trunk rotation and prolonged muscle activity may contribute to overuse complaints, particularly involving the back, shoulders and lower limbs [13–15].

Prevention strategies in windsurfing should consider both the athlete and the environment. Gradual skill progression, appropriate equipment selection, physical conditioning, warm-up routines, fatigue management and awareness of weather and water conditions may reduce injury risk [13,14]. Because windsurfing is strongly influenced by wind and wave conditions, safe participation depends not only on physical preparation, but also on matching the athlete's technical ability to environmental demands [13,14].

Kitesurfing

Kitesurfing has a distinct injury profile because the athlete is exposed to wind-generated traction forces, board speed, jumps, aerial maneuvers and potential loss of kite control. Compared with swimming, kitesurfing injuries are more commonly acute and traumatic. Mechanisms include falls, high-speed crashes, uncontrolled dragging, failed jumps, hard landings and collisions with the water surface, equipment or surrounding obstacles [15–19]. Several studies and reviews have identified the lower extremities as a particularly important injury region in kitesurfing. Foot, ankle and knee injuries may occur during twisting mechanisms, board control, failed landings or impact with the water or equipment [16–18]. In the prospective study by Nickel et al., the overall self-reported injury rate was 7 injuries per 1000 hours of practice, and the most commonly injured sites included the foot and ankle, skull, chest and knee [16]. The comparative study of windsurfing and kitesurfing at the North Sea reported injury rates of 5.2 per 1000 hours for windsurfing and 7.0 per 1000 hours for kitesurfing, with lower extremities being the most frequently affected region [15]. More recent literature has continued to emphasize the predominance of lower limb injuries, while also noting clinically important injuries involving the chest, head and trunk [17,18].

The specific demands of kitesurfing suggest that prevention should focus on technical training, progressive learning of jumps and maneuvers, safe launching and landing, proper use of safety-release systems, weather assessment and avoidance of conditions that exceed the athlete's skill level [16,19]. Equipment familiarity and supervised instruction are particularly important, as inadequate kite control and changing wind conditions may increase the risk of traumatic injury [16,19].

Wingfoiling and foiling-related windsurfing disciplines

Wingfoiling and foiling-related disciplines, including iQFOiL, represent newer and rapidly developing forms of water sports. Compared with swimming, surfing and kitesurfing, the available literature remains limited, but early studies suggest that these disciplines may have

specific injury mechanisms related to hydrofoil equipment, higher speeds, falls from an elevated board position and the technical demands of foil control [2,20,21].

In wingfoiling, injuries may occur during falls, loss of balance, failed maneuvers and contact with the board, wing or hydrofoil. The hydrofoil may represent a specific hazard because of its rigid structure and underwater position; therefore, injuries may result not only from impact with the water but also from direct contact with equipment [21]. The available wingfoiling study reported that common acute injury types included contusions, strains, cuts and abrasions, particularly involving the lower extremities [21].

In iQFOiL and other high-performance foiling windsurfing disciplines, the combination of speed, balance demands, racing intensity and technical complexity may influence injury risk in elite athletes [20]. The iQFOiL literature indicates that injuries are common among elite competitors and that many occur during on-water training, which highlights the importance of sport-specific prevention strategies and training-load monitoring [20]. Because these disciplines are relatively new, future research should focus on prospective injury surveillance, standardized injury definitions and the evaluation of protective equipment and technical training programs [2,20,21].

Sailing

Sailing is a heterogeneous sport that includes recreational cruising, offshore sailing, Olympic classes and high-performance racing. Injury patterns differ according to boat type, crew role, level of participation, weather conditions and race or voyage context. Unlike swimming, where overuse mechanisms dominate, sailing injuries are often related to the interaction between the sailor, boat, equipment and environment [22–25].

Common injury mechanisms include falls on deck, impact with the boom, contact with ropes or hardware, sudden boat movements, high-speed maneuvers and loss of balance [22–25]. Offshore and transoceanic sailing may additionally involve prolonged exposure to weather, fatigue, limited access to medical care and cumulative physical demands during long-distance voyages [22]. In recreational offshore sailors, injuries and health issues have been reported as relatively common, although severe injuries appear to be less frequent [22]. Smaller boats and limited experience with the specific boat have been identified as factors associated with a higher occurrence of injuries in this setting [22].

In competitive and Olympic sailing, injury patterns may be influenced by race intensity, repetitive maneuvers, boat class, training volume and sustained hiking positions used to counterbalance the boat during sailing [23–25]. The epidemiological review by Neville and

Folland indicated that in elite Olympic-class sailing, the lumbar and thoracic spine and the knee are commonly affected regions [23]. Surveillance during the International Sailing Federation Sailing World Championships provided additional evidence that injury and illness monitoring can identify sport- and class-specific health problems in competitive sailors [25].

Prevention in sailing should therefore be adapted to the sailing context. In offshore and recreational sailing, preventive strategies may include crew education, adequate sailing experience, first-aid training, fatigue management and safe boat handling [22]. In competitive sailing, prevention should additionally consider class-specific physical demands, conditioning, recovery, technical training and monitoring of repetitive loads affecting the spine, knees and upper limbs [23–25].

Discussion

This structured narrative review demonstrates that injuries in water sports cannot be interpreted as a single, uniform clinical category. Although the analyzed disciplines share the aquatic environment, their injury profiles are shaped by different combinations of biomechanics, equipment use, speed, environmental exposure and level of participation. The main distinction observed across the included literature is the contrast between overuse mechanisms, which are more characteristic of swimming, and acute traumatic mechanisms, which are more frequently described in board-based, wind-dependent and sailing-related sports [3–25].

Swimming represents a unique injury model among the analyzed water sports. In this discipline, injuries are mainly related to repetitive movement patterns, high training volume and cumulative biomechanical load rather than direct trauma. The shoulder is particularly vulnerable because of the repetitive overhead nature of swimming and the high number of stroke cycles performed during training [4,6–8]. However, the literature also indicates that swimming injuries should not be reduced only to shoulder disorders. Lumbar spine complaints, upper extremity problems and lower limb injuries may also occur, depending on stroke technique, training load and individual biomechanical factors [3–7]. From a clinical perspective, this suggests that injury prevention in swimmers should be based on a comprehensive approach, including load monitoring, technique correction, shoulder and trunk strengthening, mobility work and early management of pain symptoms.

In contrast, surfing, windsurfing, kitesurfing and wingfoiling are more strongly influenced by environmental and equipment-related factors. In surfing, acute injuries are commonly associated with wave impact, falls, contact with the surfboard or fins, collision with the seabed and interactions with other surfers [9–12]. These mechanisms explain the frequent involvement

of the skin, head, face, neck and extremities in surfing-related trauma [10–12]. Nevertheless, surfing also includes repetitive components, particularly paddling and prolonged prone positioning, which may contribute to overuse symptoms in the shoulder, cervical spine and lumbar spine [10,11]. Therefore, surfing should be understood as a mixed-risk sport, combining acute trauma with gradual-onset complaints.

Windsurfing and kitesurfing demonstrate how wind, speed, technical skill and equipment control influence injury risk. Windsurfing injuries may result from falls, impact with the board or sail, failed maneuvers and sudden changes in wind conditions, while repetitive sail handling and sustained stance may also contribute to overuse complaints [13–15]. Kitesurfing appears to have a more distinctly traumatic profile, with injury mechanisms related to traction forces, jumps, hard landings, uncontrolled dragging and loss of kite control [15–19]. The lower extremities, especially the foot, ankle and knee, are repeatedly emphasized as important injury sites in kitesurfing, although injuries to the trunk, chest, head and upper extremities may also occur [15–19]. These findings underline the importance of technical instruction, gradual progression, weather assessment and proper use of safety systems in wind-dependent board sports.

The emergence of wingfoiling and foiling-related windsurfing disciplines introduces additional considerations. Compared with swimming, surfing or kitesurfing, the evidence base for these sports is still limited, but available studies suggest that hydrofoil equipment, higher speeds, falls from an elevated board position and the technical difficulty of foil control may create distinct injury mechanisms [2,20,21]. The rigid structure and underwater position of the hydrofoil may represent a specific equipment-related hazard. At the same time, iQFOiL and other high-performance foiling disciplines require high levels of balance, strength, endurance and technical control, particularly in competitive athletes [20,21]. Because these disciplines are relatively new and rapidly developing, prospective injury surveillance and standardized reporting are especially needed.

Sailing differs from board sports because the athlete interacts not only with water and weather, but also with a complex boat environment. Injuries may result from falls on deck, impact with the boom, contact with ropes or hardware, sudden boat movements and high-speed maneuvers [22–25]. In offshore and transoceanic sailing, prolonged exposure, fatigue, limited access to medical care and environmental stressors may further influence injury and health risk [22]. In competitive and Olympic sailing, repetitive maneuvers, class-specific demands, training volume and sustained hiking positions used to counterbalance the boat during sailing may contribute to both acute and overuse injuries [23–25]. Therefore, prevention in sailing should

be adapted to the sailing context, with different priorities for recreational cruising, offshore sailing and elite competition.

A common theme across the reviewed literature is the importance of prevention strategies tailored to the specific discipline. In swimming, the priority is prevention of overuse injuries through load management, technique optimization and strengthening programs [3–8]. In surfing and aquatic board sports, prevention should include environmental awareness, safe equipment handling, physical preparation and strategies to reduce board-related trauma [9–12]. In windsurfing, kitesurfing and wingfoiling, technical training, appropriate equipment selection, protective equipment, safe progression and careful assessment of wind and water conditions appear particularly relevant [13–21]. In sailing, preventive strategies should include crew education, safe boat handling, fatigue management, first-aid preparedness and conditioning programs adjusted to boat class and level of competition [22–25].

The present review also reveals several limitations in the available literature. First, injury definitions and reporting methods differ substantially between studies, which makes direct comparison between sports difficult. Second, many studies are retrospective or survey-based, which increases the risk of recall bias and underreporting of less severe injuries. Third, exposure is not always reported consistently, and injury rates are therefore difficult to compare across disciplines. Fourth, newer sports such as wingfoiling and iQFOiL remain underrepresented in the literature compared with swimming, surfing, kitesurfing and traditional sailing. Finally, the available studies often combine recreational and competitive athletes, although injury mechanisms and risk profiles may differ substantially between these groups.

Future research should focus on prospective injury surveillance using standardized definitions and exposure-based injury rates. This is particularly important for newer water sports and foiling disciplines, where the rapid development of equipment and increasing popularity may change injury patterns over time. Studies comparing recreational and elite athletes would also be valuable, as prevention strategies should be adjusted to the level of participation, technical demands and environmental exposure. In addition, future work should evaluate the effectiveness of sport-specific prevention programs, including conditioning, technical training, protective equipment and safety education.

These findings have practical implications for clinicians, coaches and athletes, as prevention and rehabilitation strategies should be adapted to the biomechanical, technical and environmental demands of each discipline.

Conclusions

Water sports are associated with diverse injury patterns that depend on the specific discipline, level of participation, equipment use and environmental conditions. Swimming is predominantly related to overuse injuries, especially involving the shoulder, lumbar spine and upper extremity, whereas surfing, windsurfing, kitesurfing, wingfoiling and sailing are more frequently associated with acute traumatic mechanisms related to falls, collisions, speed, wind, waves and contact with equipment.

The reviewed literature indicates that prevention strategies should be tailored to the demands of each sport. In swimming, prevention should focus mainly on training load management, technique optimization, strengthening and early recognition of overuse symptoms. In board-based, wind-dependent and sailing-related sports, prevention should additionally emphasize technical training, environmental risk assessment, safe equipment use, protective measures, fatigue management and first-aid preparedness.

Future studies should use standardized injury definitions, prospective surveillance methods and exposure-based injury reporting to improve comparability between water sports. Further research is particularly needed for newer and rapidly developing disciplines, such as wingfoiling and foiling-related windsurfing classes.

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